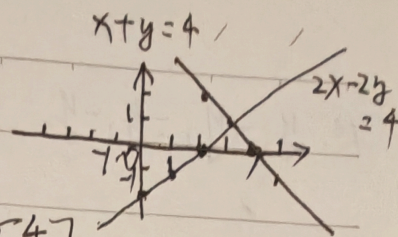


1.2

1. row picture:



column picture: $x \begin{bmatrix} 1 \\ 2 \end{bmatrix} + y \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \end{bmatrix}$

3. I think the intersection is a line.

If $u=-1$, then $\begin{cases} u=-1 \\ v=2 \\ w=3 \\ z=2 \end{cases}$

The equation $u+v+w=3$ can make the equations have no solution.

$$\begin{cases} u+v+w=2 & (1) \\ u+2v+3w=1 & (2) \\ v+2w=0 & (3) \end{cases}$$

$(1) - (2) + (3) \Rightarrow 0=1$, so there is no solution.

The last zero should be replaced by -1

one of the solutions can be $\begin{cases} u=1 \\ v=2 \\ w=-1 \end{cases}$

$$10. y_2 - y_1 = y_3 - y_2$$

20. point

We can know that:
$$\begin{cases} x &= a \\ x+y &= b \\ x+y+z &= c \\ x+y+z+t &= d \end{cases}$$

20. point

We can know that:
$$\begin{cases} x+y+z+t=3 \\ y+z+t=3 \\ z+t=3 \\ t=2 \end{cases} \Rightarrow \begin{cases} x=0 \\ y=0 \\ z=1 \\ t=2 \end{cases}$$

1.3

1. $f=5$,
$$\begin{cases} 2x+3y=1 \\ -6y=6 \end{cases}$$

4. $f = \frac{c}{a}$; the second pivot is $d - \frac{bc}{a}$

$$y = \frac{g - \frac{cf}{a}}{\frac{ad-bc}{a}} = \frac{ag-cf}{ad-bc}$$

10.

$$10. \begin{cases} 2x + 3y + z = 8 \\ \cancel{2y + 2z = 4} \end{cases} \quad 10. \begin{cases} 2x + 3y + z = 8 \\ ① y + 3z = 4 \\ ② z = 8 \end{cases}$$

The solution is: $\begin{cases} z = 1 \\ y = 1 \\ x = 2 \end{cases}$

18. ~~(b)~~ Solutions represent the intersection of some planes. So if the planes can intersect at two points, then they can intersect at a line, which means there ~~is~~ should be other solutions.

(a) If (x, y, z) and (X, Y, Z) are two solutions, then $(\frac{x+X}{2}, \frac{y+Y}{2}, \frac{z+Z}{2})$ should also be a solution.

$$23. \begin{cases} u + v + w = 2 \\ 2v + 2w = -2 \\ 2w = 2 \end{cases} \Rightarrow \begin{cases} u = 3 \\ v = -2 \\ w = 1 \end{cases}$$